

Mathematical Definition of Resonance by Tyrone Mineard

The Resonance Architecture models interaction as a governed dynamical system in which behavioral state selection, interaction alignment, and governance constraints jointly determine effective system behavior. Rather than treating response generation as a single inference step, the architecture represents interaction as a continuously evolving state whose coherence can be measured, evaluated, and regulated over time.

This mathematical framework is built upon five fundamental concepts:

- Interaction State
- Behavioral Basin Selection
- Resonance Gain
- Governance Damping
- Effective Resonant Output

Together, these constructs provide a formal description of how interaction stability emerges and is maintained throughout the interaction process.

1. Interaction State Vector

At any moment in time, the interaction between participating cognitive agents is represented by an interaction state vector.

$$\mathbf{S}(\mathbf{t}) = [\sigma(\mathbf{t}), \tau(\mathbf{t}), \pi(\mathbf{t}), \nu(\mathbf{t}), \kappa(\mathbf{t})]^T$$

where:

- $\sigma(\mathbf{t})$ represents semantic coherence.
- $\tau(\mathbf{t})$ represents tone alignment.
- $\pi(\mathbf{t})$ represents persona stability.
- $\nu(\mathbf{t})$ represents narrative continuity.
- $\kappa(\mathbf{t})$ represents the symbolic control state.

Rather than describing a single response, the interaction state vector characterizes the complete state of the interaction field at a given instant. Each element captures a distinct dimension of interaction quality, while the vector as a whole represents the current behavioral condition of the system.

2. Behavioral Basin Selection

Interaction does not evolve within a single behavioral mode. Instead, the system operates within one of several stable behavioral regions, referred to as **behavioral basins**. These

basins correspond to persistent operating modes such as reasoning strategies, personas, interaction policies, or symbolic configurations.

The active behavioral basin is selected according to:

$$**B^*(X) = \arg \max_i [w_i \phi_i(X)]^{**}$$

where:

- **X** is the current interaction context.
- $\phi_i(X)$ is the compatibility score between the current context and behavioral basin i .
- w_i is a weighting coefficient representing the relative preference or importance of basin i .
- $**B^*(X)^{**}$ is the selected behavioral basin.

The compatibility function may be implemented using learned classifiers, embedding similarity metrics, policy networks, or other scoring mechanisms depending on the underlying implementation. The selected basin establishes the behavioral mode that governs subsequent response generation.

3. Resonance Gain

Once a behavioral basin has been selected, the quality of interaction is amplified through the combined effects of alignment, coupling strength, and signal clarity.

The resonance gain is defined as:

$$G = A \times C \times S$$

where:

- **A** represents interaction alignment between participating agents.
- **C** represents coupling strength, describing the responsiveness and adaptability of the interaction.
- **S** represents signal clarity or communication quality.

Higher values of the resonance gain indicate stronger interaction coherence and more effective information transfer between participants.